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CHEMISTRY

Grade 12

Paper 1 Multiple Choice

May 2018

1 hour

Candidates answer on the Question Paper.

Additional Materials: Calculator
 Data Booklet

12CHEM/01

READ THESE INSTRUCTIONS FIRST

Write your centre number and candidate number in the spaces at the top of the page.

Write in dark blue or black pen.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct by putting a tick (✓) in the correct box.

Each correct answer will score one mark.

Any rough working can be done in this booklet.

A Data Booklet is provided.

The total number of marks for this paper is 40.

For Examiner's Use

Total

This document consists of **17** printed pages and **3** blank pages.

Section A

For
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Use

For each question there are four possible answers, **A**, **B**, **C**, and **D**. Choose the one you consider to be correct by putting a tick (✓) in the correct box.

1 Which model of the atom first included the existence of electron shells?

A Ernest Rutherford's model

B J J Thomson's model

C John Dalton's model

D Niels Bohr's model

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

2 Element 41, niobium, forms several compounds with oxygen and chlorine. Analysis of one of these compounds by mass shows it to be 37.55% niobium, 43.05% chlorine and 19.40% oxygen.

What is the empirical formula of this compound?

A $\text{Nb}_2\text{Cl}_2\text{O}$

B $\text{Nb}(\text{ClO})_3$

C $\text{Nb}(\text{ClO})_5$

D NbClO_3

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 3 If neutrons are fired at ^{10}B atoms the nuclei can each capture a neutron and undergo fission leading to two new nuclei, **Q** and **R**.

Nucleus **Q** is the nucleus of a noble gas atom, which consists of equal numbers of protons and neutrons.

What is nucleus **R**?

- A** ^7Be
B ^8Be
C ^6Li
D ^7Li

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 4 Which statement about **all** the electrons in a nitrogen molecule is correct?

- A** There are one bond pair and eight lone pairs.
B There are one bond pair and six lone pairs.
C There are three bond pairs and four lone pairs.
D There are three bond pairs and two lone pairs.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 5 The table contains statements about the chlorides of beryllium and barium and suggested reasons for the statements.

Which row of the table is correct?

	statement	reason
A	BaCl_2 is more covalent than BeCl_2	Ba has a greater effective nuclear charge than Be
B	BaCl_2 is more covalent than BeCl_2	Be has a greater effective nuclear charge than Ba
C	BeCl_2 is more covalent than BaCl_2	Ba has a greater effective nuclear charge than Be
D	BeCl_2 is more covalent than BaCl_2	Be has a greater effective nuclear charge than Ba

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

- 6 Chlorine reacts separately with cold water and with an excess of cold dilute aqueous sodium hydroxide.

Which equation is correct for **one** of the two reactions that takes place?

- A $\text{Cl}_2 + \text{H}_2\text{O} \rightarrow \text{HCl} + \text{HOCl}$
 B $2\text{Cl}_2 + 2\text{H}_2\text{O} \rightarrow 4\text{HCl} + \text{O}_2$
 C $\text{Cl}_2 + \text{NaOH} \rightarrow \text{HCl} + \text{NaOCl}$
 D $3\text{Cl}_2 + 6\text{NaOH} \rightarrow 5\text{NaCl} + \text{NaClO}_3 + 3\text{H}_2\text{O}$

A ☐ B ☐ C ☐ D ☐

[1]

- 7 Which statement is correct?

- A The conversion of Sn^{2+} ions to Sn^{4+} ions is oxidation.
 B The formation of bromide ions from bromine atoms is oxidation.
 C When bromine is made from hydrogen bromide, bromine is reduced.
 D When SnO_2 is made from SnO , tin is reduced.

A ☐ B ☐ C ☐ D ☐

[1]

- 8 Which equation can be derived from the ideal gas equation?

- A $\frac{p}{R} = nTV$
 B $pVT = nR$
 C $\frac{RT}{pV} = \frac{1}{n}$
 D $R = \frac{nT}{pV}$

A ☐ B ☐ C ☐ D ☐

[1]

9 Which statement about the **ammonia molecule** is correct?

- A** Lone-pair bond-pair repulsion is greater than bond-pair bond-pair repulsion.
B Lone-pair lone-pair repulsion is greater than lone-pair bond-pair repulsion.
C The bond angle is greater than the bond angle in a methane molecule.
D The bond angle is smaller than the bond angle in a water molecule.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

10 Oxides of nitrogen in the atmosphere and nitrate compounds in rivers can have an environmental impact.

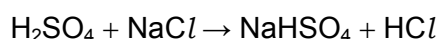
Which statement is true?

- A** Low levels of nitrate compounds are toxic to water plants.
B Nitrate compounds combine with oxygen and this removes oxygen from river water.
C Nitrogen dioxide, NO_2 , will oxidise sulfur dioxide in the atmosphere.
D Nitrogen oxide, NO , will oxidise sulfur dioxide in the atmosphere.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

11 Concentrated sulfuric acid reacts with sodium chloride crystals.



Which row of the table is correct for this reaction?

	concentrated sulfuric acid behaves as an acid	concentrated sulfuric acid behaves as an oxidising agent
A	no	no
B	no	yes
C	yes	no
D	yes	yes

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

12 Which statement explains why carbon monoxide, CO, is toxic?

- A CO molecules prevent haemoglobin molecules from transporting oxygen to body tissues.
- B CO molecules prevent the oxidation of haemoglobin molecules by oxygen.
- C CO molecules prevent the transfer of nitrogen molecules away from body tissues.
- D CO molecules reduce Fe^{3+} ions in haemoglobin to Fe^{2+} ions.

A ☐ B ☐ C ☐ D ☐

[1]

13 Propanone reacts with iodine. The reaction is acid catalysed. The rate equation for this reaction is shown.

$$\text{rate} = k[\text{propanone}][\text{H}^+]$$

A reaction mixture is produced in which the concentrations of propanone, H^+ ions and iodine are all $0.0250 \text{ mol dm}^{-3}$. The initial rate of the reaction is $0.0840 \text{ mol dm}^{-3} \text{ s}^{-1}$.

What is the numerical value of k , correct to three significant figures, under these conditions?

- A 1.31×10^{-6}
- B 5.25×10^{-5}
- C 134
- D 5380

A ☐ B ☐ C ☐ D ☐

[1]

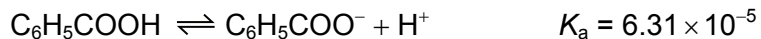
14 Which equation has an enthalpy change equal to the lattice enthalpy of sodium sulfide?

- A $\text{Na}^+(\text{g}) + \text{S}^-(\text{g}) \rightarrow \text{NaS}(\text{g})$
- B $\text{Na}^+(\text{g}) + \text{S}^-(\text{g}) \rightarrow \text{NaS}(\text{s})$
- C $2\text{Na}^+(\text{g}) + \text{S}^{2-}(\text{g}) \rightarrow \text{Na}_2\text{S}(\text{g})$
- D $2\text{Na}^+(\text{g}) + \text{S}^{2-}(\text{g}) \rightarrow \text{Na}_2\text{S}(\text{s})$

A ☐ B ☐ C ☐ D ☐

[1]

- 15 Benzoic acid, $\text{C}_6\text{H}_5\text{COOH}$, is a weak acid.



What is the $\text{p}K_b$ of the benzoate ion, $\text{C}_6\text{H}_5\text{COO}^-$?

- A 1.58×10^{-10}
 B 4.2
 C 8.8
 D 9.8

A ☐ B ☐ C ☐ D ☐

[1]

- 16 A 25.00 cm^3 sample of a solution of strontium hydroxide is titrated against $0.0400 \text{ mol dm}^{-3}$ hydrochloric acid. The endpoint is reached when exactly 37.25 cm^3 of acid has been added.

What is the concentration of the original solution of strontium hydroxide?

- A $0.0134 \text{ mol dm}^{-3}$
 B $0.0298 \text{ mol dm}^{-3}$
 C $0.0596 \text{ mol dm}^{-3}$
 D $0.119 \text{ mol dm}^{-3}$

A ☐ B ☐ C ☐ D ☐

[1]

- 17 Structural isomerism and stereoisomerism should be taken into account when answering this question.

How many isomers exist with molecular formula C_4H_8 ?

- A 3
 B 4
 C 5
 D 6

A ☐ B ☐ C ☐ D ☐

[1]

18 Structural isomerism **only** should be taken into account when answering this question.

How many isomeric primary alcohols exist with the molecular formula $C_5H_{11}OH$?

- A 2
- B 3
- C 4
- D 5

A ☐ B ☐ C ☐ D ☐

[1]

19 The fractionation of oil produces many useful fractions.

Which statement is correct?

- A Diesel is used as a fuel by many small boats.
- B Fuel oil is used as aviation fuel.
- C Gasoline is used for domestic heating.
- D Refinery gas is used by most road vehicles.

A ☐ B ☐ C ☐ D ☐

[1]

20 Ethanol is an important industrial chemical. It is synthesised in large quantities.

What is one method for the large-scale synthesis of ethanol by the chemical industry?

- A enzymatic fermentation of proteins
- B hydration of ethene in the presence of a catalyst
- C hydrolysis of ethyl ethanoate in the presence of a catalyst
- D reduction of ethanal

A ☐ B ☐ C ☐ D ☐

[1]

- 21 If 2-bromopropane is treated with a solution of potassium hydroxide two types of reaction are possible. The two types of reaction are substitution and elimination.

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Which conditions of the potassium hydroxide solution **most** favour elimination?

	solvent	temperature and dilution
A	ethanol	cold, dilute
B	ethanol	hot, concentrated
C	water	cold, dilute
D	water	hot, concentrated

A ☐ B ☐ C ☐ D ☐

[1]

- 22 Chromium, cobalt, copper and lead are all metals that can be extracted from the Earth's crust.

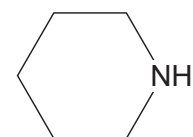
Which metal is **least** likely to be found in the Earth's crust in the native state?

- A chromium
B cobalt
C copper
D lead

A ☐ B ☐ C ☐ D ☐

[1]

- 23 Compound X is heterocyclic.



compound X

What is the name of compound X?

- A piperidine
B pyridine
C pyrrole
D pyrrolidine

A ☐ B ☐ C ☐ D ☐

[1]

24 A propanone molecule has one functional group.

Which row of the table is correct?

	name of functional group	bonding in functional group
A	carbonyl	two π bonds
B	carbonyl	one π bond and one σ bond
C	carboxyl	two π bonds
D	carboxyl	one π bond and one σ bond

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

25 Which statement about benzene is correct?

- A** Benzene reacts with ethanoyl chloride in an acylation reaction; no catalyst is used.
- B** Ethanoyl chloride behaves as a nucleophile when it reacts with benzene.
- C** The carbon to carbon bonds in benzene all have the same bond length.
- D** When benzene is hydrogenated to form cyclohexane the reaction is endothermic.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

26 When compound **Z** is treated with warm Tollens' reagent, no visible change is seen.

When compound **Z** is treated with sodium metal, no visible change is seen.

What could be compound **Z**?

- A** pentanal
- B** pentan-2-ol
- C** pentanoic acid
- D** pentanone

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

27 A chemist makes a polymer using alanine as the only monomer.

Which bonds or linkages form between neighbouring alanine monomer residues?

- A amide linkages
- B C–C bonds
- C C–O bonds
- D ester linkages

A ☐ B ☐ C ☐ D ☐

[1]

28 A chemist takes 7.80 g of benzene and treats it with a nitrating mixture.

He distils out product **Q**. Product **Q** has one nitrogen atom per molecule.

He then adds tin metal and concentrated HCl to his sample of product **Q** and heats it under reflux for one hour. He then neutralises the mixture and distils out 2.21 g of product **R**.

What is the percentage yield of product **R** from the 7.80 g of benzene?

- A 17.1%
- B 18.0%
- C 23.8%
- D 28.3%

A ☐ B ☐ C ☐ D ☐

[1]

- 29 Succinate dehydrogenase is an enzyme. It catalyses the oxidation of succinate to fumarate in human cells.

Malonic acid is an inhibitor of this process. This inhibition can be reversed at high succinate concentrations.

Which statement is true?

- A Malonic acid binds to fumarate and prevents it leaving the active site of succinate dehydrogenase.
- B Malonic acid binds to succinate and prevents it binding to the active site of succinate dehydrogenase.
- C Malonic acid binds to succinate dehydrogenase but **not** at the enzyme's active site.
- D Malonic acid binds to the active site of succinate dehydrogenase.

A ☐ B ☐ C ☐ D ☐

[1]

- 30 The DNA molecule is double stranded.

How are the two strands of DNA held together?

- A by electrostatic forces between water molecules and negatively charged phosphate groups
- B by hydrogen bonds between polar organic bases
- C by hydrogen bonds between the deoxyribose residues
- D by van der Waals forces between non-polar organic bases

A ☐ B ☐ C ☐ D ☐

[1]

Section B starts on the next page.

Section B

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For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

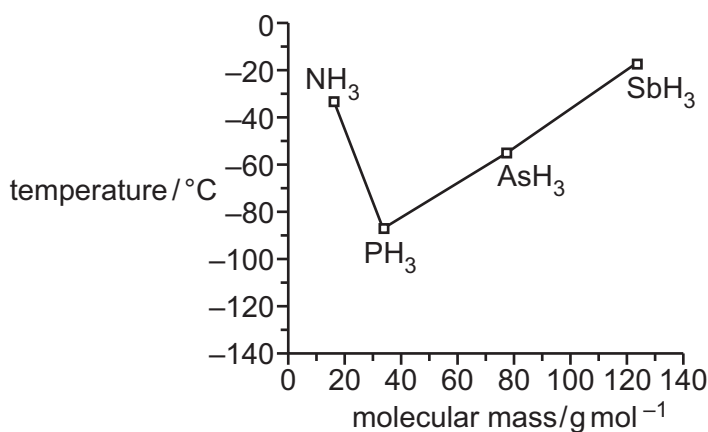
Decide whether each of the statements is correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

No other combination of statements is used as a correct response.

31 The graph shows the boiling points of the hydrides of the Group 15 elements.



The boiling points can be explained by referring to the intermolecular forces in the hydrides.

Which statements are correct?

- 1 SbH₃ has the strongest hydrogen bonding so it has the highest boiling point.
- 2 NH₃ has hydrogen bonding, which gives it a higher boiling point than PH₃.
- 3 AsH₃ has a higher boiling point than PH₃ because it has stronger induced dipole–dipole forces.

A ☐ B ☐ C ☐ D ☐

[1]

32 What is involved in a mass spectrometer?

- 1 a magnetic field
- 2 an electric field
- 3 an evacuated tube or chamber

A ☐ B ☐ C ☐ D ☐

[1]

33 The Periodic Table can be used to make predictions about the properties of different elements.

Which predictions about germanium and bromine can be made on the basis of their positions in the Periodic Table?

- 1 Germanium has a lower first ionisation energy than bromine.
- 2 Oxides of germanium are less acidic than oxides of bromine.
- 3 Germanium has a lower melting point than bromine.

A ☐ B ☐ C ☐ D ☐

[1]

34 The element sulfur exists in various forms called allotropes.

Which statements about allotropes of sulfur are correct?

- 1 α -sulfur consists of rhombic crystals.
- 2 β -sulfur consists of monoclinic crystals.
- 3 Plastic sulfur consists of cubic crystals.

A ☐ B ☐ C ☐ D ☐

[1]

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

No other combination of statements is used as a correct response.

35 Iron and iron compounds have many uses as heterogeneous and homogeneous catalysts.

The table shows three important chemical reactions and their catalysts.

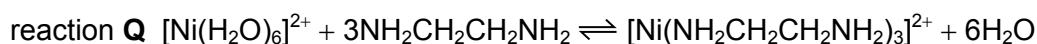
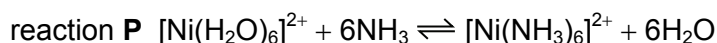
In which reactions are the catalysts **heterogeneous**?

	reaction	catalyst
1	$\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$	Fe
2	$\text{C}_2\text{H}_4 + \text{Cl}_2 \rightarrow \text{CH}_2\text{ClCH}_2\text{Cl}$	FeCl_3
3	$\text{CO} + \text{H}_2\text{O} \rightleftharpoons \text{CO}_2 + \text{H}_2$	Fe_3O_4

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

36 Ammonia, NH_3 , is a monodentate ligand. Ethylene diamine, $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$, is a bidentate ligand. Both ligands form a complex with Ni^{2+} ions, as shown in reactions **P** and **Q**.



The $[\text{Ni}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$ is much more stable than the $[\text{Ni}(\text{NH}_3)_6]^{2+}$ complex.

Which statements help to explain the difference in the stabilities of the two ions?

- 1 Reaction **Q** has a much more positive ΔS than reaction **P**.
- 2 $[\text{Ni}(\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2)_3]^{2+}$ is trigonal planar; $[\text{Ni}(\text{NH}_3)_6]^{2+}$ is octahedral.
- 3 Reaction **Q** has a much more positive ΔG than reaction **P**.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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- 37** If sodium hydroxide solution is added to bromomethane a nucleophilic substitution reaction occurs.

Which steps occur in this reaction?

- 1 homolytic fission of a C–Br covalent bond
- 2 formation of a C–O covalent bond
- 3 attack by an electron pair on a carbon atom with a partial positive charge

A ☐ B ☐ C ☐ D ☐

[1]

- 38** Polymer **Y** is biodegradable.

Which monomers might be used to make polymer **Y**?

- 1 leucine and valine
- 2 ethane-1, 2-diol and propanedioic acid
- 3 propene and *cis* but-2-ene

A ☐ B ☐ C ☐ D ☐

[1]

- 39** A sample of serine is dissolved in a buffer solution at pH 2.0.

Which species will be present in the solution?

- 1 $\text{H}_3\text{N}^+\text{-CH}(\text{CH}_2\text{OH})\text{-COOH}$
- 2 $\text{H}_3\text{N}^+\text{-CH}(\text{CH}_2\text{OH})\text{-COO}^-$
- 3 $\text{H}_2\text{N-CH}(\text{CH}_2\text{OH})\text{-COO}^-$

A ☐ B ☐ C ☐ D ☐

[1]

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct.	1 and 2 only are correct.	2 and 3 only are correct.	1 only is correct.

No other combination of statements is used as a correct response.

- 40** A scientist is studying the bacterium *E. coli*. She knows that a particular length of DNA, called a gene, codes for the production of protein **A**. Protein **A** is involved in the transportation of glucose into the bacterial cells.

She causes a mutation in the gene, so that one particular guanosine residue (G) is changed into a cytosine residue (C).

What **could be** the result of this mutation?

- 1 The bacterium can no longer transport glucose into its cells.
- 2 The bacterium can transport glucose into its cells more efficiently.
- 3 The bacterium is unaffected.

A ☐ **B** ☐ **C** ☐ **D** ☐

[1]

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